

Data Visualization and Exploration

COMS7056A/COMS4060A

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Lecture 2



Admin updates

- Moodle
- Discord



Lesson Plan

- Descriptive statistics
- Plots and visualisations
- Data cleaning
- Missing data
- Outliers

Descriptive Statistics

- A way to organise and summarise data.
- Mean, variance, standard deviation etc.
- To understand your data to some extent.
- Useful for getting quick overview of the data.
- However...
 - Using descriptive stats can hide the shape of distribution
 - Data is not usually normal.
 - Distribution can be skewed.
 - Missing Values/outliers/special values.
 - Still want to see some stats first.

Some Descriptive Stats

- Count: number of values per category.
- Mean: the average
- Mode: most frequent value
- Median: middle value, divides dataset in two
- Percentiles: position in a dataset, nth percentile has n% of terms less than this.
- Variance: (std dev)^2
- Range: min to max
- Standard deviation (std dev):
 - Population:

$$\sigma = \sqrt{\frac{1}{N} \sum_{i=0}^N (x_i - \mu)^2}$$

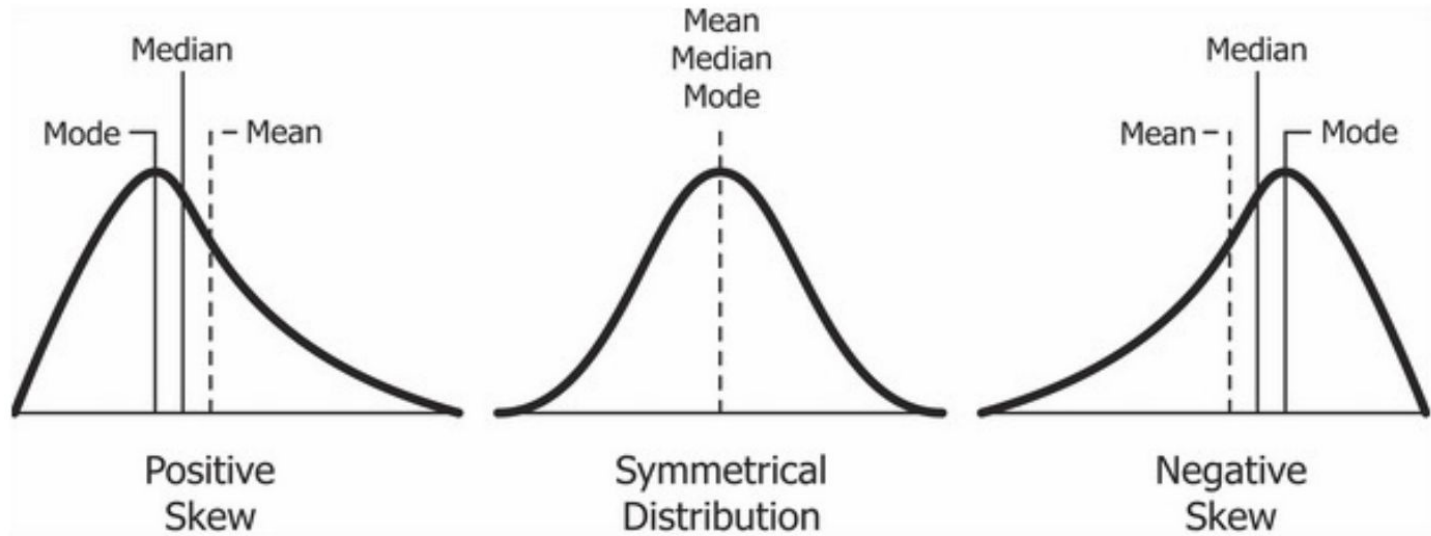
Sample:

$$s = \sqrt{\frac{1}{N-1} \sum_{i=0}^N (x_i - \bar{x})^2}$$

Skewness

- Measure of asymmetry of a distribution about the mean.
- Skewness is positive, negative or undefined.
- In normal distribution, tails in each side are exactly the same.
- Distribution skewed to the left (negative skewness) has tail on the LHS longer than on the RHS, ie. the mean $<$ mode.
- Distribution skewed to the right (positive skewness) has tail on the RHS longer than on the LHS, ie. the mean $>$ mode

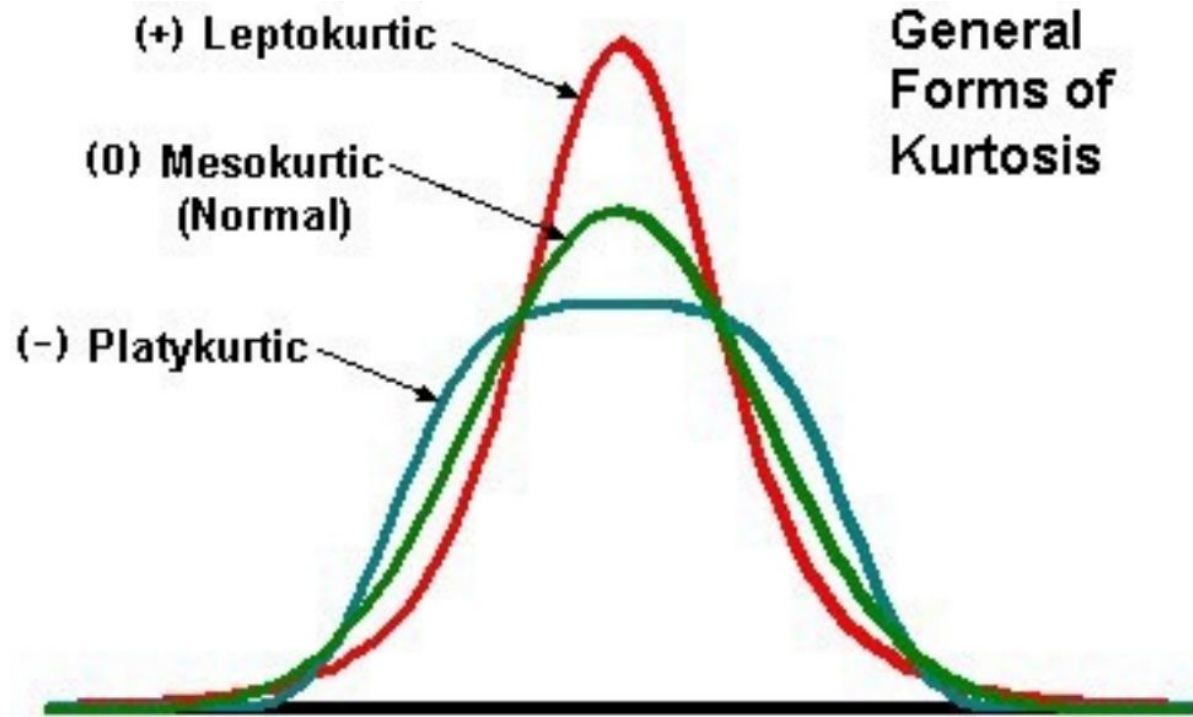
Skewness



Kurtosis

- Measure of whether there is a heavy-tailed (presence of outliers) or light-tailed (lack of outliers) distribution relative to a normal distribution.
- Mesokurtic
 - Distribution with a similar kurtosis to a normal distribution kurtosis, ie, 0.
- Leptokurtic
 - Positive kurtosis
 - Thick/heavy tailed distribution.
 - If more peaked than Mesokurtic curve, called Leptokurtic curve.
- Platykurtic
 - Negative kurtosis
 - Thinner tails than normal distribution
 - If less peaked than Mesokurtic curve, called Platykurtic curve

Kurtosis



Skewness Vs Kurtosis

- The difference between **skewness** and **kurtosis** is that the skewness shows the degree of symmetry, and kurtosis refers to the degree of outliers.

Correlation

- Indication of how strongly two (or more) variables are related.
- We look at the correlation coefficient to quantify this:
 - Value ranges between -1 and +1
 - The larger the absolute value of the corr coeff, the stronger the relationship
 - $(0, 1]$: there is a positive correlation, meaning if variable A goes up, so does variable B
 - 0: variable A does not influence variable B in any way
 - $[-1, 0)$: if variable A goes up, variable B goes down

Correlation

- Pearson's Coefficient: the most common, identifies linear relationships.
- Spearman's Coefficient: quantifies how well the relationship between variables can be described with a monotonic function. Can be used to find non-linear relationship
- Kendall's Tau: similar to Spearman's, used for testing dependence of variables

Penguin Dataset

```
df.head()
```

	species	island	bill_length_mm	bill_depth_mm	flipper_length_mm	body_mass_g	sex
0	Adelie	Torgersen	39.1	18.7	181.0	3750.0	Male
1	Adelie	Torgersen	39.5	17.4	186.0	3800.0	Female
2	Adelie	Torgersen	40.3	18.0	195.0	3250.0	Female
3	Adelie	Torgersen	NaN	NaN	NaN	NaN	NaN
4	Adelie	Torgersen	36.7	19.3	193.0	3450.0	Female

```
df[['species', 'island']].value_counts()
```

species	island	
Gentoo	Biscoe	124
Chinstrap	Dream	68
Adelie	Dream	56
	Torgersen	52
	Biscoe	44

dtype: int64

```
df.describe()
```

	bill_length_mm	bill_depth_mm	flipper_length_mm	body_mass_g
count	342.000000	342.000000	342.000000	342.000000
mean	43.921930	17.151170	200.915205	4201.754386
std	5.459584	1.974793	14.061714	801.954536
min	32.100000	13.100000	172.000000	2700.000000
25%	39.225000	15.600000	190.000000	3550.000000
50%	44.450000	17.300000	197.000000	4050.000000
75%	48.500000	18.700000	213.000000	4750.000000
max	59.600000	21.500000	231.000000	6300.000000

Iris Dataset

```
dfi.head()
```

	sepal_length	sepal_width	petal_length	petal_width	species
0	5.1	3.5	1.4	0.2	setosa
1	4.9	3.0	1.4	0.2	setosa
2	4.7	3.2	1.3	0.2	setosa
3	4.6	3.1	1.5	0.2	setosa
4	5.0	3.6	1.4	0.2	setosa

```
dfi.groupby('species').describe()
```

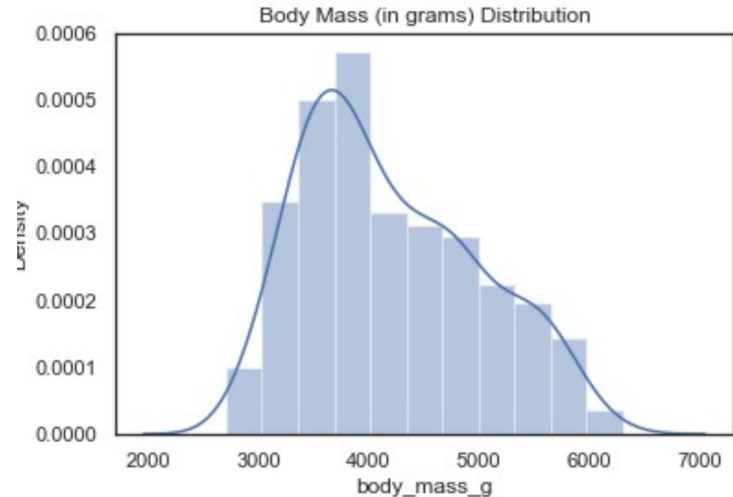
	sepal_length								sepal_width			...	petal_length												petal_width					
	count	mean		std	min	25%	50%	75%	max	count	mean	...	75%	max	count	mean		std	min	25%	50%	75%	max							
species																														
setosa	50.0	5.006	0.352490	4.3	4.800	5.0	5.2	5.8	50.0	3.428	...	1.575	1.9	50.0	0.246	0.105386	0.1	0.2	0.2	0.3	0.6									
versicolor	50.0	5.936	0.516171	4.9	5.600	5.9	6.3	7.0	50.0	2.770	...	4.600	5.1	50.0	1.326	0.197753	1.0	1.2	1.3	1.5	1.8									
virginica	50.0	6.588	0.635880	4.9	6.225	6.5	6.9	7.9	50.0	2.974	...	5.875	6.9	50.0	2.026	0.274650	1.4	1.8	2.0	2.3	2.5									

```
dfi.describe()
```

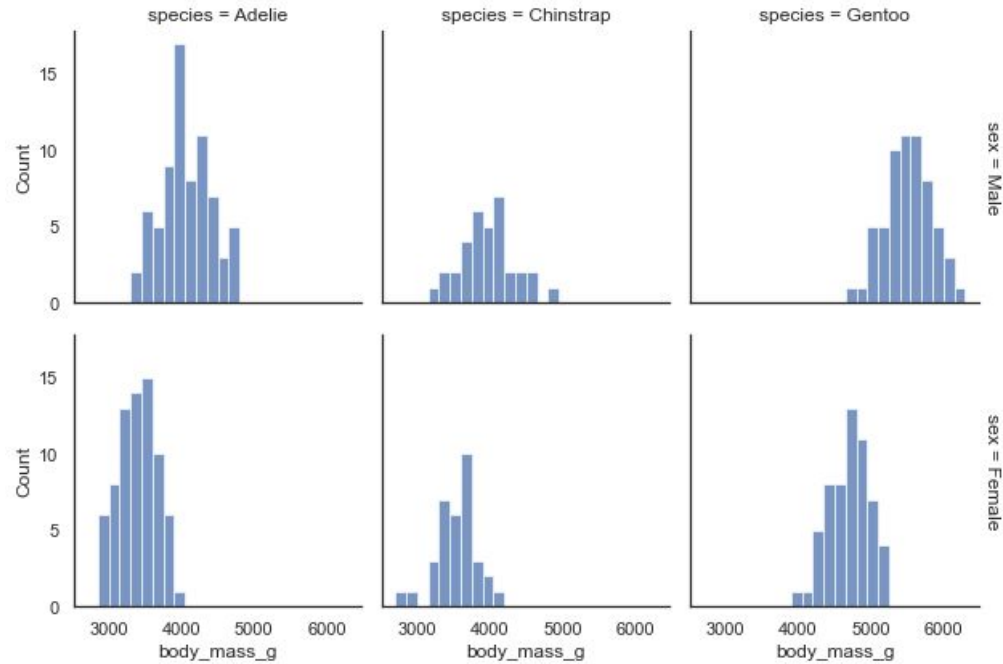
	sepal_length	sepal_width	petal_length	petal_width
count	150.000000	150.000000	150.000000	150.000000
mean	5.843333	3.057333	3.758000	1.199333
std	0.828066	0.435866	1.765298	0.762238
min	4.300000	2.000000	1.000000	0.100000
25%	5.100000	2.800000	1.600000	0.300000
50%	5.800000	3.000000	4.350000	1.300000
75%	6.400000	3.300000	5.100000	1.800000
max	7.900000	4.400000	6.900000	2.500000

Plots

- Visualisations offer a quick, intuitive, and simple way of conveying critical concepts.
- Descriptive stats make it difficult to view relationships between different variables.
- Eg. Body mass in penguins:
 - Negative kurtosis
 - Mean of 4200, median 4050, mode 3800,
 - Min 2700, max 6300
 - Vis is much easier to interpret

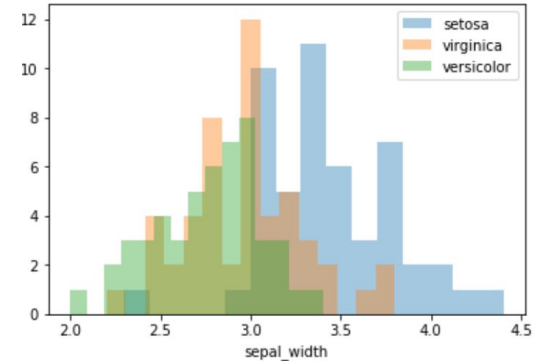
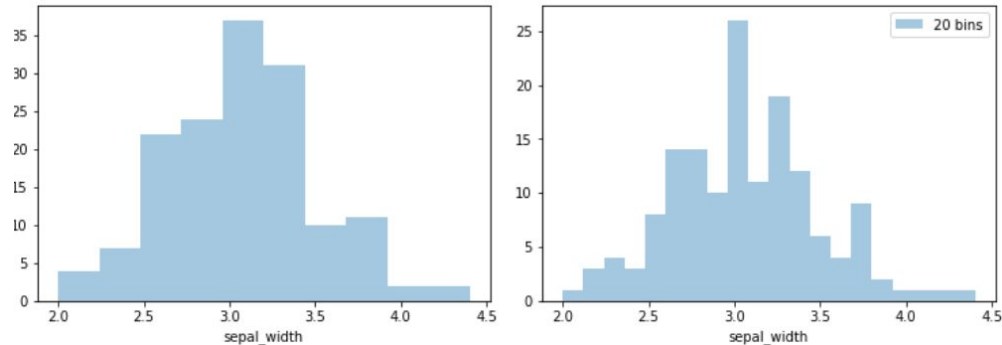


Plots



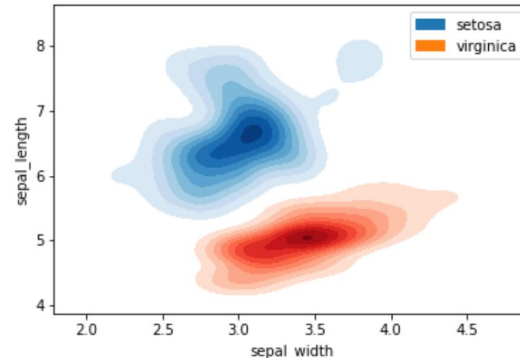
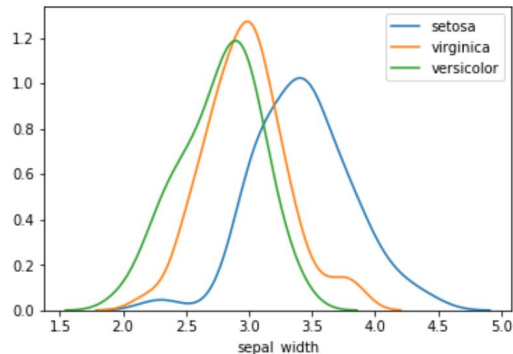
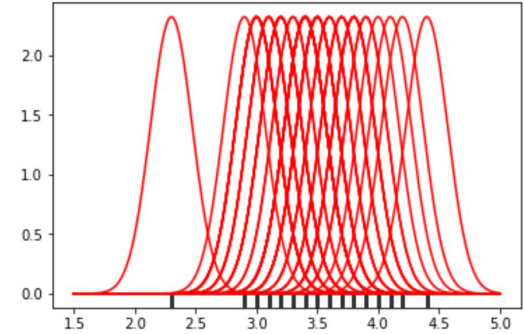
Histograms

- An approximate view of the distribution for single continuous variable.
- Divide the variable into bins, count how many data points are in each bin, and plot the bin intervals vs counts.
- This can help to smooth out the data.
- Choosing the bin width is the hard part! Really an art more than science...
Need to play around to get it right.



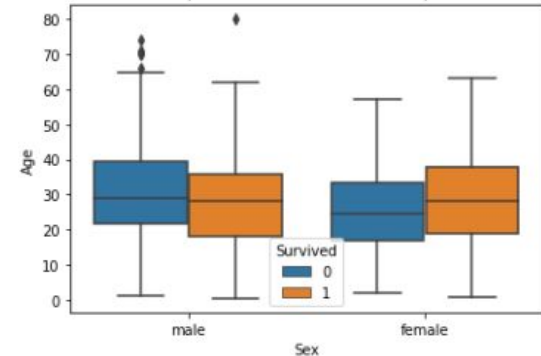
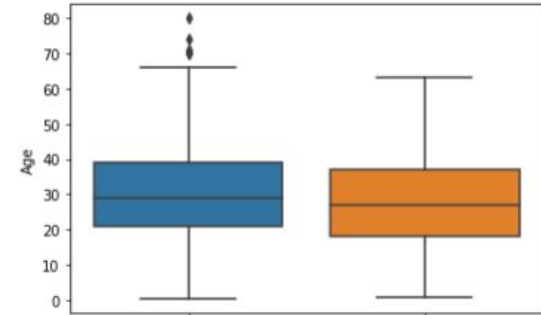
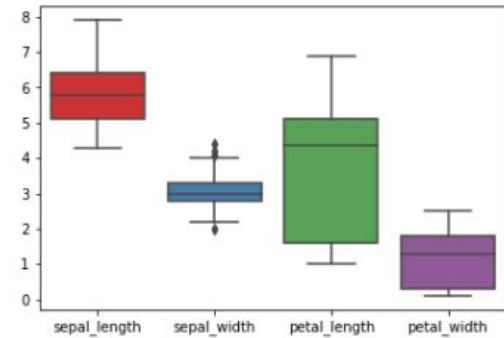
Kernel Density Estimation (KDE)

- Essentially a more complex histogram
- Every data point is replaced by a continuous curve
- These curves are all added together at the end
- Gaussian kernel
- The result is a smoothed density estimation.
- The y-axis is the probability density (NOT a probability)
- useful when trying to compare variables on the same axis



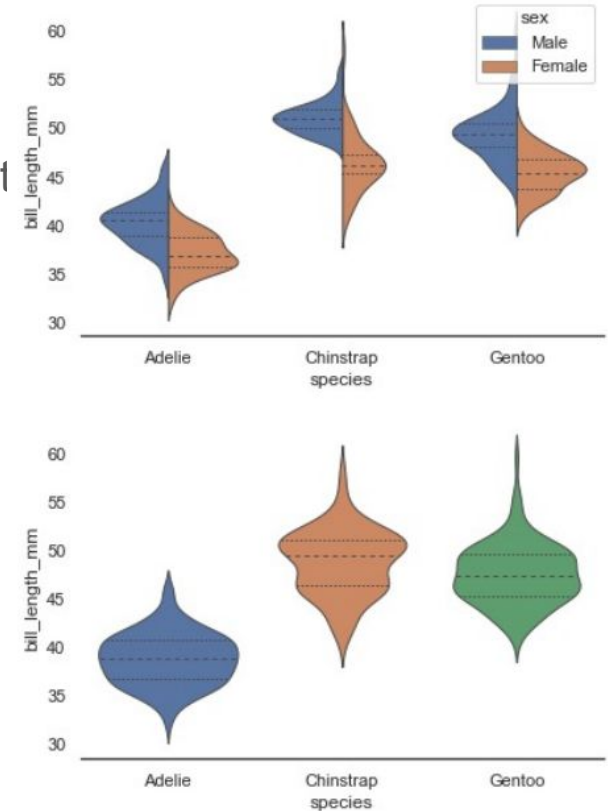
Boxplots

- A quick way to visualise the data based on five descriptive statistical measures:
 - Min and max (whiskers)
 - Median
 - First and third quartile
- The whiskers can be used to represent one standard deviation from the mean, or the top and bottom percentiles
 - This is great for highlighting outliers
 - Anything data outside this is represented as a dot or circle



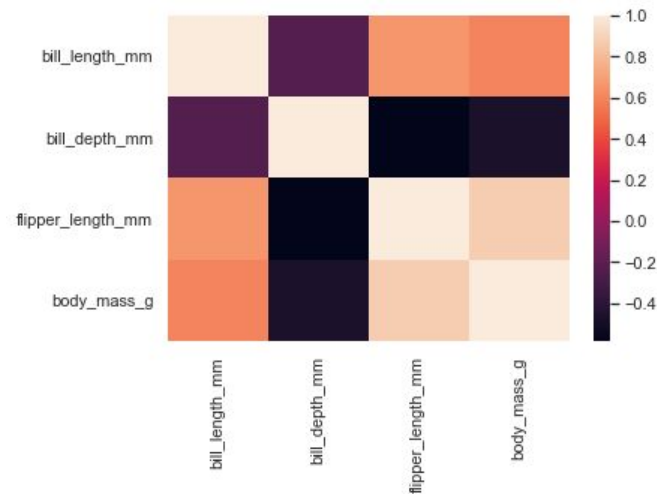
Violin Plots

- Similar to boxplots, except that there is a kernel density plot added along the vertical
- Allows a view of all the data, not only median, mode, etc.
 - Especially useful for multimodal data

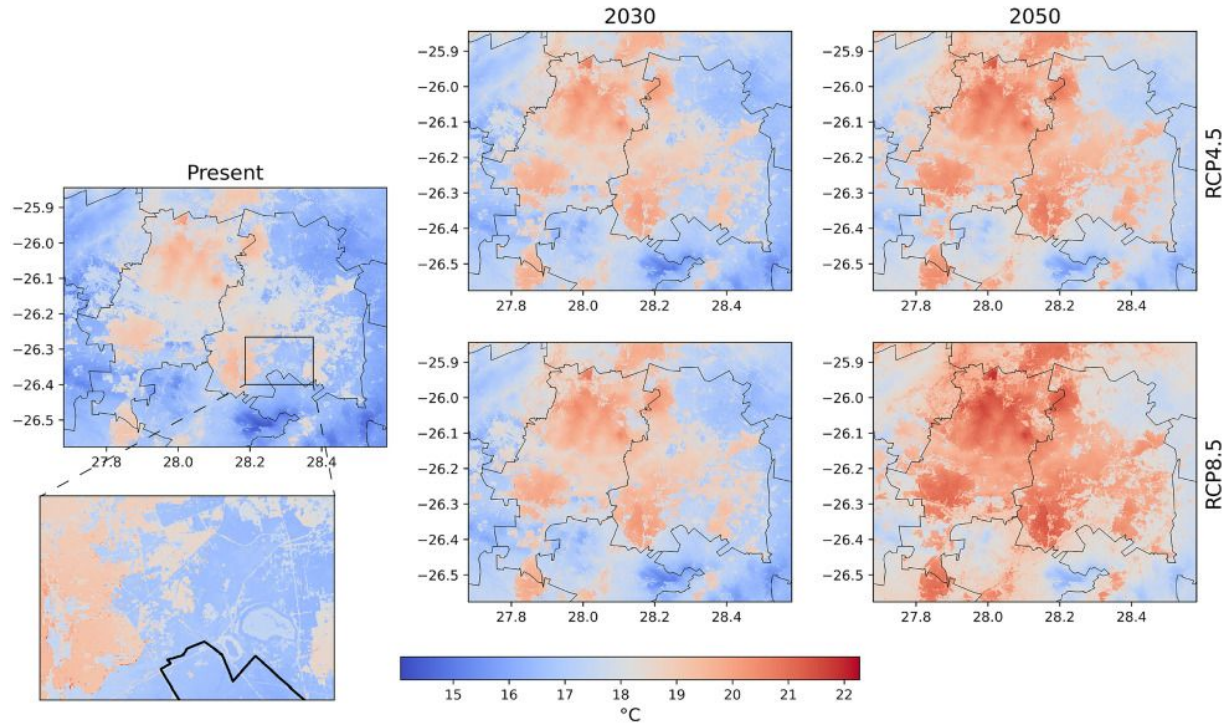


Heatmaps

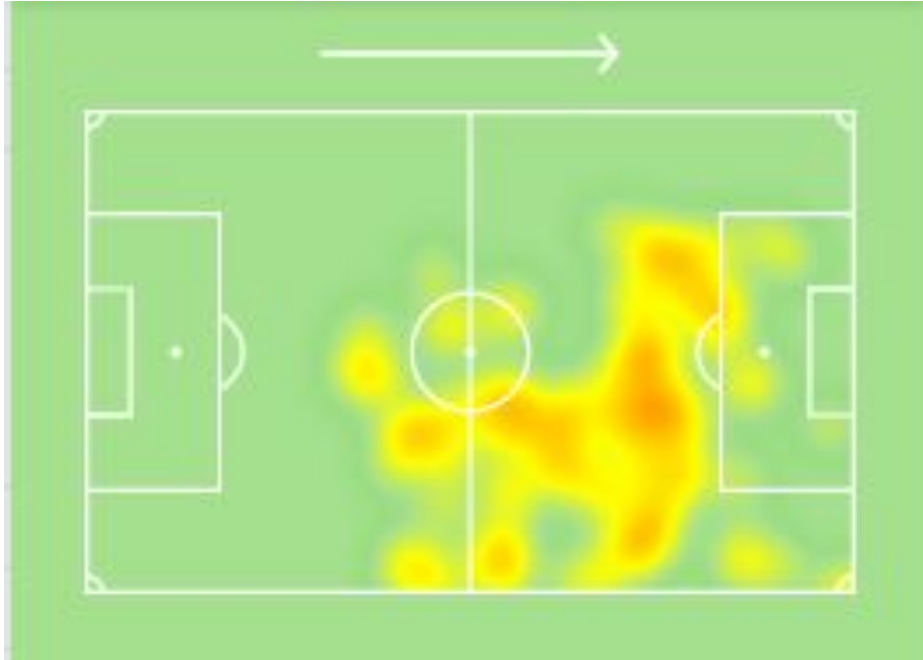
- Heatmaps use colour to display the magnitude of values in a dataset
 - The colour changes by hue to indicate larger or lower values
- a matrix is displayed with rows and columns that indicate discrete values or categories and the relationships between them
 - Cluster: In particular, this is often used for correlation tables or showing missing values
 - Spatial: the colour indicates the magnitude of values in the specific location



Heatmaps

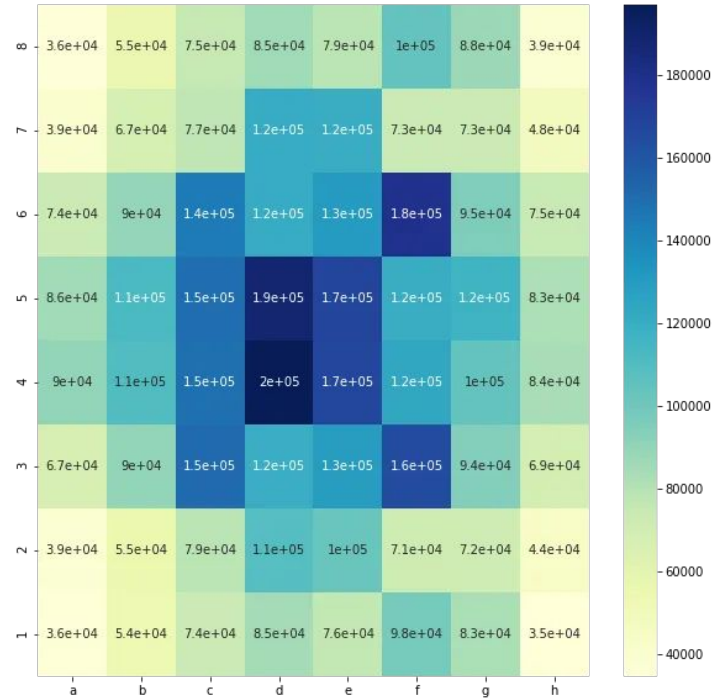


Heatmaps



<https://barcauniversal.com/messis-heat-map-against-osasuna-is-in-the-shape-of-a-goat/>

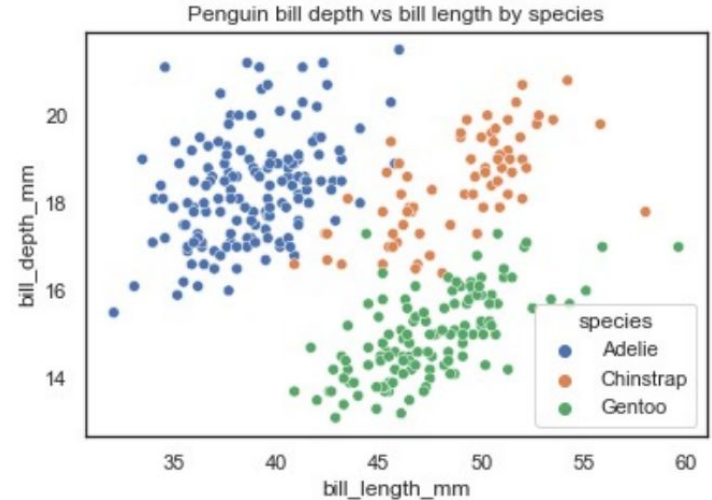
Heatmaps



<https://www.kaggle.com/datasets/rishidamarla/chess-games>

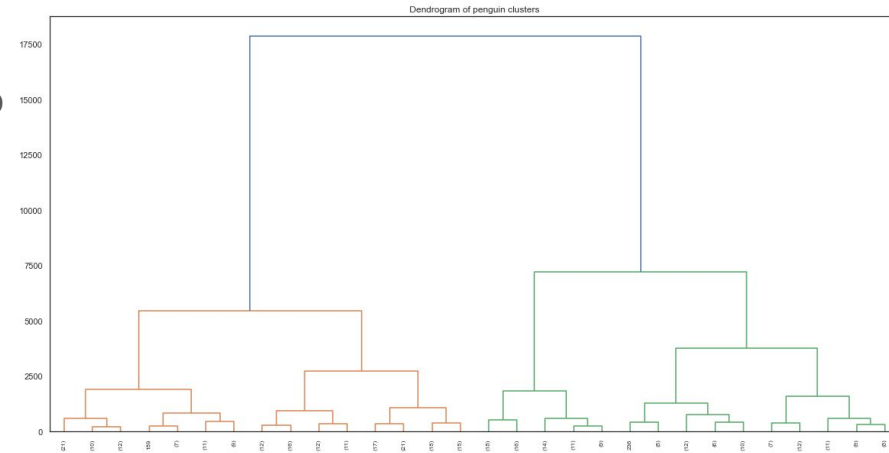
Scatter Plots

- Scatter plots are useful to get a view of the relationship between two variables
 - A third variable can be added as categorical variable.
- Don't interpret **correlation as causation**.
 - Observing a relationship between two variables in a scatter plot does not mean that changes in one variable are responsible for changes in the other.
 - Possible that the observed relationship is due to some additional variable that affects both plotted variable, or it is coincidental



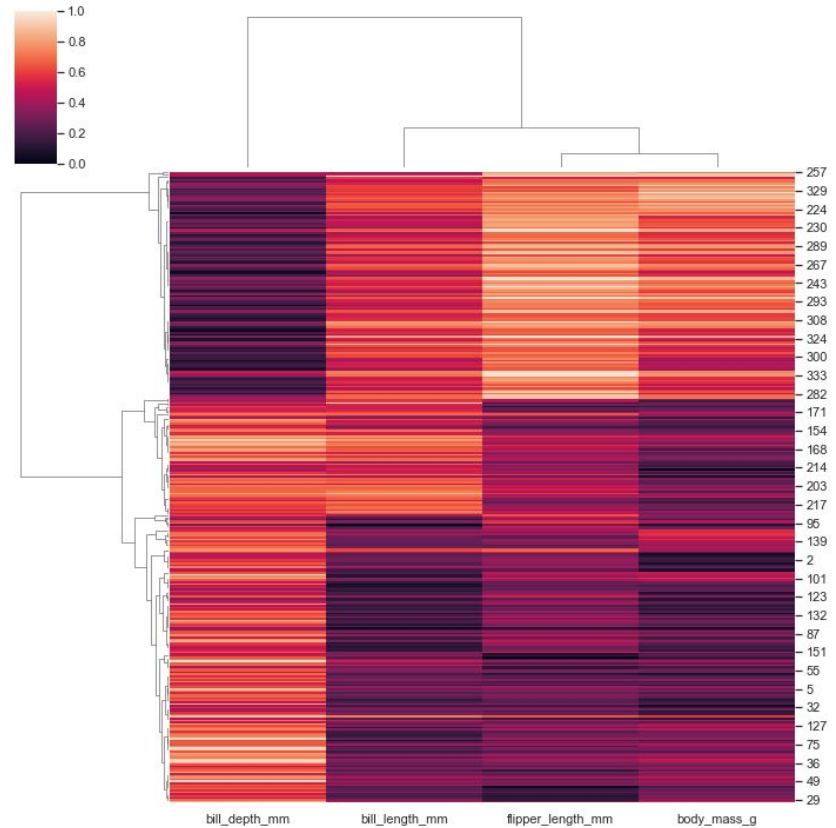
Dendrograms/cluster maps

- Dendrograms show the hierarchical connection between the objects.
 - Dendrograms are used to figure out a path to allocate objects to clusters
 - A distance metric is used to group the data, shown on the y-axis. Lower values (horizontal bars) indicate the greater similarity.
 - X-axis refer to explicit entries in the dataset



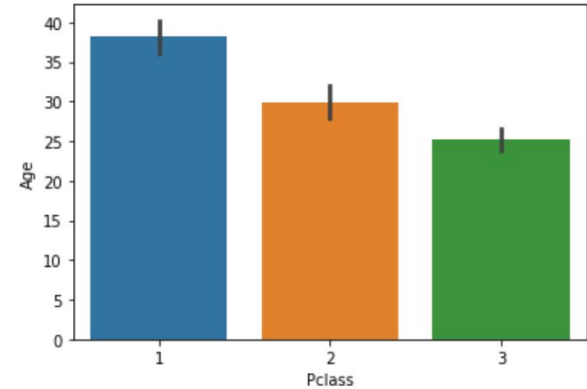
Dendrograms/cluster maps

- Cluster maps show the relationship between two (or more) variables
 - It shows a dendrogram that shows similar behaviour



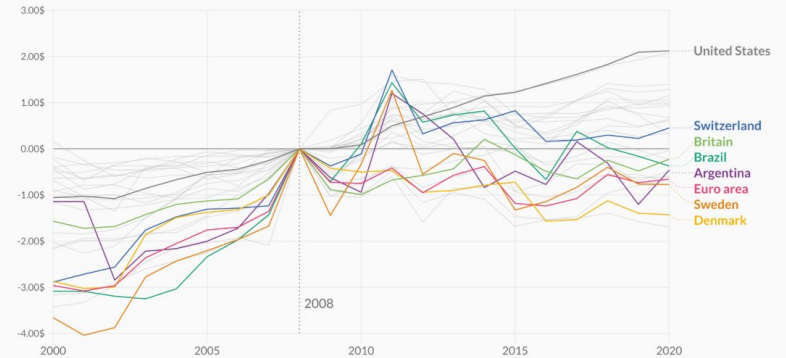
Charts

- Bar/line/pie
 - Bar/pie: These types of charts are used to compare the quantities/counts of different categories.
 - Also useful to show multiple categories and error bars
- Line: It is used to plot the relationship of dependence of one variable on another like if you want to show data over very long periods or continuously changing data, ie, time series data



Compared to the financial crisis in 2008, how much more or less do you have to pay for a Big Mac today?

The index chart visualizes the price changes (in USD) of a Big Mac based on a 2008 as index year. The Big Mac Index is published by The Economist as an informal way to provide a test of the extent to which market exchange rates result in goods costing the same in different countries. It seeks to make exchange-rate theory a bit more digestible and takes its name from the Big Mac, a hamburger sold at McDonald's restaurants.

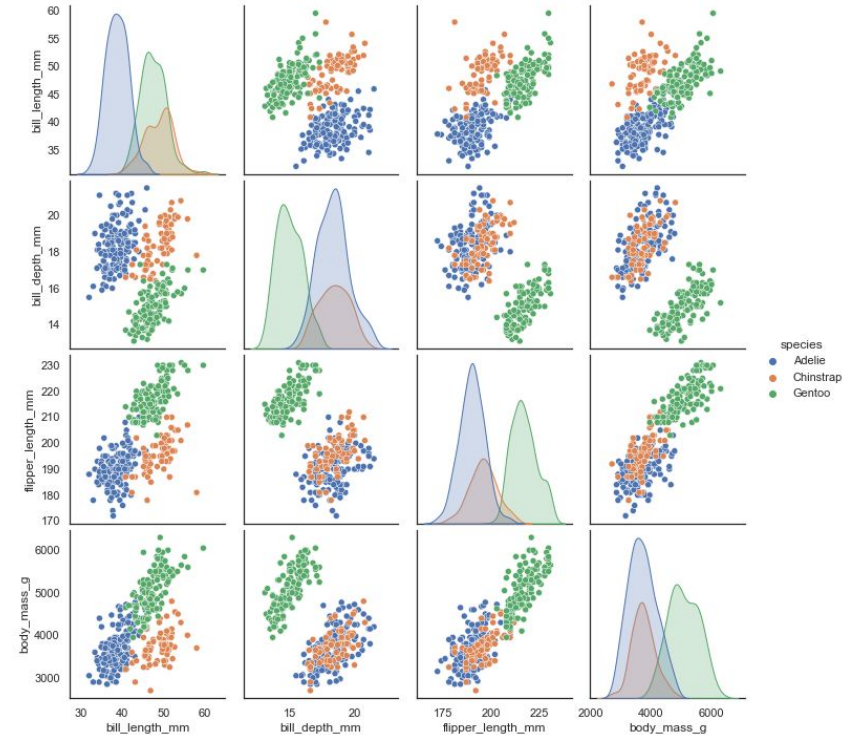


Visualization by Cédric Scherer • Data by The Economist • The index chart shows the 27 countries that provide Big Mac prices for all years from 2000 to 2020. In case a country was reported twice per year, the mean value was visualized.



Putting it all together for Penguins

- Flipper length and bill length are linearly separable
- Gentoo is different in all measurements
- Clustering is probably going to work on a number of these relationships



When to use which kind of plot?

- Univariate Analysis
 - Used for exploring a single variable.
 - Numerical and categorical variables (eg. Text) handled differently.
- Bi/multivariate Analysis
 - Used to explore the relationship between 2+ different variables

	Univariate		Bivariate/Multivariate analysis			
	Numerical	Categorical	Numerical vs categorical	Numerical X2 + categorical	Numerical vs numerical	Categorical + Categorical
Scatter plot				X	X	
Cluster map						X
Heatmap						X
Pie chart		X				
Histogram	X					
Distplot	X		X			
Boxplot/violin plot	X		X			
Bar chart			X			
Dendrograms						X

Data Cleaning

- **Missing values:** Datasets can contain missing values... how do we handle this?
- **Special values:** Numeric variables can have special values including $\pm\text{Inf}$, NA and NaN. Calculations involving these values often result in nonsensical results, or, they need special treatment. We need to handle this or clean it up.
- **Outliers:** Some (small) subset of the data that falls far outside the distribution of the rest of the values. These should be detected, but we don't always want to remove them. This should be a statistical decision.
- **Obvious inconsistencies:** These will require some domain knowledge, but some are general – make sure ages aren't negative, underage people don't have drivers licenses, a client has not driven 100 000 km in one month. We need to find the inconsistencies and plan for them.

Data Cleaning

- **Data integrity:** Are there duplicate values in the data? This is often an issue when doing joins on multiple tables in SQL, for example.
- **Important terms:**
 - **Bias:** tendency for over or under estimation of some parameter. High bias means too simple and does not have enough info giving poor accuracy and underfitting
 - **Variance:** variability in the data/predictions. High variance means the model is paying too much attention to the training data and cannot generalise



Data Integrity & Inconsistencies

- **Duplicate observations**
 - This normally happens when combining data from multiple sources (or from multiple departments in a company)
 - Scraping data from websites
 - Running queries on a database
- To identify duplicates, it's a good idea to try a group by on specific columns followed by a count (some packages have a duplicate finding method built-in)
 - These should be dropped from the dataset!
- **Irrelevant observations** are those that don't apply the specific problem that we're trying to solve.
 - For example, if you were building a model for Single-Family homes only, you wouldn't want observations for Apartments in there.
 - Specifically, look at charts of the categorical features in your dataset and look for classes that don't make sense
 - Do this before engineering features
- **Mislabeled classes**, i.e. separate classes that should really be the same or typos
 - E.g. If 'N/A' and 'Not Applicable' appear as two separate classes, you should combine them.
 - E.g. 'IT' and 'information_technology' should be a single class.
 - E.g. Labels of "Pothole", "pothole" and "potholes" should all be grouped together
 - Use bar charts to look for this. Of course this doesn't work if you have many categories – in which case you might want to try some kind of fuzzy matching to identify duplicates

Missing Values

- Missing values can have a serious impact on the analysis, but it does not always mean that the analysis is doomed
- It's important to look at the cases where there is missing data and figure out if there is a sensible root cause for this.
- There are three general types of scenarios for this:
 - Missing Completely At Random (MCAR),
 - Missing At Random (MAR),
 - Missing Not At Random (MNAR)
- **MCAR**: the missing values do not have any correlation with any other variables, observed or not.
- **MAR**: there are missing values in the data, but the cause is not completely random – it can be explained, either by observed or unobserved data
- **MNAR**: exclusive of the above, and refers to the case where the true value of the missing data is directly affected by the reason it is missing. The missing value is telling you something about the dataset

Imputation

- How do we handle these cases?
 - If it's MCAR, then deletion might be okay, but it will decrease the sample size and decrease the power of the analysis
- Mean/median substitution: fill in a constant value, such as the mean
 - There might also be multiple sub-populations in the dataset that have different means
 - It removes any correlation with this variable and others
 - However, does not change the sample mean for that specific variable
 - Using the *median* is often better
 - A big problem with this is it affects any modelling outcomes

Imputation

- Regression: Imputation by regression/ ANOVA. Essentially build a regression model from the rest of the variables, and use that model to predict the missing values. However, this does not allow for any variance in the value
- Hot-Deck Imputation: Find other entries that have similar other variables, and randomly choose one of their values to fill in the missing feature.
- Cold-Deck Imputation: The idea here is to use a separate dataset from which to draw the missing data. This can be done in a random or systematic way.

```
run_classifier(df, 'Original') # original dataset
run_classifier(df_miss, 'Missing Removed') # remove missing data (note this cuts the dataset in half!)
run_classifier(df_mean, 'Mean Imputation') # impute with the mean
run_classifier(df_med, 'Median Imputation') # impute with the median
```

Original Accuracy: 0.763

Missing Removed Accuracy: 0.781

Mean Imputation Accuracy: 0.762

Median Imputation Accuracy: 0.762

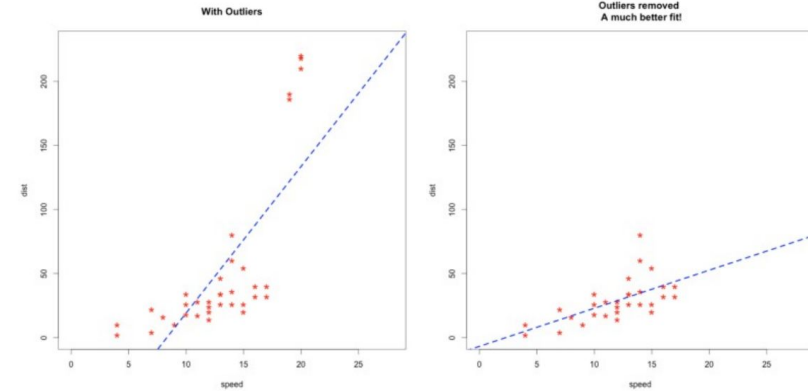


Special Values

- Similar to missing values, but there are actually values, they are just not necessarily meaningful or physical
- Examples include:
 - NaN (Not a Number): this is usually indicative of a missing value. Something important to note here is that some aggregation functions (count, sum, etc) might actually just omit these values completely – check for this!
 - -1, +-999: This often gets used to indicate an error or missing value too. It will generally be seen only when this is a value that cannot be a physical or reasonable value. Eg an unknown age, might be filled with a negative value to indicate that it is missing
 - +-inf: Either there was a divide by zero error, or it really does mean a large number – it's worth looking into whether a div by zero is possible, otherwise, consider capping the value at some large number (that is not infinity)
- Note: the second two examples should be noticeable when looking for outliers!

Outliers

- Outliers can influence the training of the model, or even the visualisations.
 - Does not necessarily mean it should be discarded!
 - Example is where you are looking for a rare event.
 - Sometime outliers can be discarded, especially if it does not seem a likely outcome.
 - Removing it reduces the variance, but this means that other, previously “normal” values are not outliers – this is referred to as “masking”. This can be performed multiple times.
- In multivariate data it's more difficult – this will depend on the distribution of your data! We may need to apply some transformations first
 - We'll tackle this in some of the other lectures, specifically: PCA



Outliers

- Using percentiles: we can choose to remove the top and bottom X percentiles
- Scatter plots: this often gives you a quick visualization and works for 2D data
- Using standard deviation (Z-score): if the dataset has a Normal or Gaussian distribution, then we can look at how many standard deviations away from the mean a value is. We can then truncate the tails.
- Isolation forest: uses a collection of binary decision trees. Its main assumption is that outliers are rare and far removed from the other data points
- Histogram-based: useful for identifying extreme values
- Distance metric based: KNN or DBScan (another lecture)
- We can choose to either remove outliers, leave them in, or cap them at an artificial limit. Choosing when to do this will depend on the problem.
- You can find a bunch of examples of this implemented in Python here:
 - <https://www.analyticsvidhya.com/blog/2019/02/outlier-detection-python-pyod/>



Outliers Methods Overview

	Pros	Cons
Z-score	• Effective if dataset is Gaussian • Easy to implement	• Doesn't work in high dimensions • If the dataset is not parametric it is not a good fit
Nearest Neighbour	• Good when the distribution is not known • Works well in higher dimensions • Easy to visualise	• Care needs to be taken to scale correctly, as distances are important • Selection of parameters is hard • Unsupervised and may need to be calibrated for each dataset
Isolation Forests	• No scaling needed • Good when distribution is unknown • Few parameters	• Hard to visualise • Can be computationally expensive if not optimised
Histograms	• Easy to implement • Easy to understand	• Doesn't work in high dimensions